AI For Material Science (EP208n) Project:

Machine Learning Based approach to predicting material properties from composition based descriptors

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Introduction

The efficient use of pre-existing computational data is critical to the acceleration and optimization of materials discovery. This project shows how machine learning techniques may be used to predict a wide range of material properties with great efficiency and accuracy when combined with quantum mechanical computations and chemical similarity concepts. We created a methodical framework that effectively connects basic, easily accessible system characteristics to intricate material behaviors by examining one-dimensional chain systems. Rapid, high-fidelity property estimate is made possible by the use of distinct material "fingerprints" that are determined from either electron charge density distributions or chemo-structural composition. In the end, the systematic exploration of large chemical spaces made possible by the integration of these sophisticated prediction models greatly accelerates the design and discovery of new, application-specific materials.

The fundamental idea is based on an obvious principle: materials with comparable structures and chemical compositions inherently have similar physical characteristics. Researchers strictly use Density Functional Theory (DFT) to create a basic training dataset rather than performing computationally costly quantum simulations for each new candidate.

These baseline materials' atomic structure or electron density is converted into numerical "fingerprints." A machine learning approach called Kernel Ridge Regression then uses the mathematical "distance" between various structures to connect these fingerprints to certain material attributes. The model totally avoids complicated quantum equations once it has been trained. It can quickly and precisely forecast the behavior of completely unknown polymer chains, turning the slow, trial-and-error process of materials discovery into a very quick, data-driven science.

Problem Statement

1. **Vast Chemical Space**

   "Chemical space" is the theoretical sum of all potential arrangements of molecules and materials. It is regarded as "vast" because of the combinatorial explosion. If the periodic table were an alphabet, the number of possible "words" that might be created by mixing its elements in different ratios, sequences, and three-dimensional geometries would increase exponentially.

For instance, there are around 30,000 different ways to create an 8-block polymer chain just by rearranging a few fundamental building elements. The combinations become nearly endless when scaled to all potential multi-element materials, well surpassing the number of atoms in the observable universe.

Ultra-fast machine learning models are now the only practical means of navigating this space and finding new materials because it is both monetarily and technically unfeasible to synthesis or computationally simulate every possible candidate.

2. **Costly Simulations**

   Density Functional Theory and other quantum mechanical models are extremely accurate but computationally intractable. On a supercomputer, solving intricate electron interactions for a single material can take days. This enormous time and processing cost results in an unachievable discovery bottleneck when trying to screen tens of thousands of candidate items.

3. **Time-Effective Solutions**

   Conventional methods of discovery rely on rigorous quantum simulations such as Density Functional Theory or experimental trial-and-error. These simulations can take days on a supercomputer for a single material, despite their excellent accuracy.

   This slow pace results in a huge bottleneck because there are almost an endless number of conceivable chemical combinations. We urgently need high-speed machine learning models that can screen tens of thousands of candidates in a matter of seconds in order to find innovative materials for advanced applications, such as next-generation batteries or semiconductors.

Proposed ML Model

The Chemical Similarity Philosophy:

The model is based on a basic chemical intuition: materials with comparable atomic configurations will inherently display comparable macroscopic characteristics.

Rather than creating a new physics simulation for each new material, we teach an algorithm to mathematically map these similarities.

Vectorization:

Chemical names or three-dimensional atomic diagrams cannot be processed by machine learning algorithms; instead, they need highly organized numerical data.

The Solution: Physical atomic patterns are transformed into pure data points by explicitly converting each possible polymer chain into a mathematical representation called a "chemo-structural fingerprint."

The Model's Architecture

20-Dimensional Feature Space: A precise 20-component vector is created by breaking down and encoding each substance. The fraction of individual chemical blocks (monomers), neighboring pairs (dimers), and three-block sequences (trimers) inside the chain are represented mathematically by these 20 properties.

Eight Target Parameters: This 20-dimensional input is simultaneously mapped to eight different material properties (including structural, electrical, and thermodynamic characteristics).

175 Data Points: The fundamental training universe is based on precisely 175 extremely exact Density Functional Theory (DFT) computations. This fundamental collection is the only source from which the model learns the laws of physics.

The Kernel Ridge Regression Predictive Engine:

The approach plots the 20-dimensional fingerprint of a completely unknown material after it has been vectorized.

Methodologies And Tools

Kernal Regression Model

Kernel regression is a type of non-parametric algorithm used for estimating the expected value of a random variable. While linear regression uses a predetermined form of estimation, such as a straight line, kernel regression can be seen as more flexible, since it allows the model to fit itself based on its own calculations, such as the averaging of nearby points.Kernels are used to determine how much weight should be given to each point in respect to their distance from *x*, and are usually of Gaussian or Epanechnikov kind, which are bell and parabola shapes, respectively. The only crucial hyperparameter is the bandwidth *(h)*, which represents whether the kernel is too tight or too loose in smoothing the trend. When the bandwidth is small, there will be overfitting; when large, underfitting.

In this project, we specifically use the Gaussian Kernel Regression method to find our results, which is given as:

$$K_h(x, x_i) = \frac{1}{h\sqrt{2\pi}} \exp\left\{-\frac{(x-x_i)^2}{2h^2}\right\}$$

h **=** bandwidth of the function

x **=** training data set points

The final formula emerging from this Gaussian Kernel function is:

$$\hat{m}_h(x) = \frac{\sum_{i=1}^{n} K_h(x - x_i) y_i}{\sum_{i=1}^{n} K_h(x - x_i)}$$

Y_i = training data set points

Density Field Theory

Density Functional Theory (DFT) is a quantum mechanical computational technique employed to study the electronic structure of many-particle systems such as atoms, molecules, and condensed matter phases. Rather than tackling the impossible problem of solving the many-electron wavefunction *Ψ,* DFT sidesteps this issue by demonstrating that all ground state properties can be derived from the electron density function *ρ(r).* The theoretical framework underlying Density Functional Theory is based on the Hohenberg-Kohn theorem that shows how the ground state energy is uniquely dependent on the electron density. However, in order to simplify the problem, the Kohn-Sham equations introduce the idea of replacing the real physical system with a hypothetical one of non-interacting electrons within an effective local potential.

The exact formula of DFT to be used in our model is given by:

$$r = \frac{\sum_{i=1}^{n}\left(X_i - \bar{X}\right)\left(Y_i - \bar{Y}\right)}{\sqrt{\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2}\sqrt{\sum_{i=1}^{n}\left(Y_i - \bar{Y}\right)^2}}.$$

Scikit-Learn

Scikit-learn is the premier Python package for machine learning, offering users straightforward and effective methods to implement predictive analytics. The scikit-learn package has a number of supervised and unsupervised algorithms built into it, such as support vector machines, decision trees, random forests, k-means clustering, and gradient boosting.

The key strength of this package lies in its intuitive API that allows for easy preprocessing, modeling, and evaluation. Scikit-learn has a wide range of built-in functions to handle issues such as cross-validation and feature extraction, allowing one to design robust ML pipelines without too much trouble.

Pdfplumber

Pdfplumber is a unique Python package that specializes in extracting information from PDFs in a very thorough manner. Unlike other PDF parsers that read PDF files as plain texts, pdfplumber considers the PDF file to be made up of geometric shapes. As such, the package enables users to obtain metadata for each individual character, line, and rectangle on the document.

The best part about this package is its robust table parser, which automatically detects cell edges in PDF files that lack any cell borders. Such a capability makes pdfplumber an essential package for data engineers working on extracting information from complicated financial and legal documents and converting them to data frames or JSONs.

Working Of The Model

1. 1D Polymer Chains

The chains in this project are 1D polymers, which act as the basis of our mathematical model. Real-world polymers exhibit irregularities at their ends, which add complexity to their chemistry. The simplicity of our mathematical model allows us to circumvent the difficulties posed by "end-group effects," enabling us to study the intrinsic properties of the basic structural units of the material.

We were able to create a wide range of 1D chains using CH2 groups and halides from Group IV. Importantly, infinite 1D structures are significantly easier to simulate using DFT than real-world 3D materials. This makes it possible for us to quickly obtain high-quality data, the essential "ground truth" needed for training our machine learning algorithms.

2. Creation of Datasets

In order to construct this database, the researchers made use of an extremely accurate method called Density Functional Theory (DFT), which is an advanced computational tool based on quantum mechanics. Using DFT, the exact physical and electronic characteristics of 175 specifically chosen 4-block polymers were calculated. In doing so, a detailed mathematical model was constructed that could relate each particular atomic configuration (CH_2 and Group IV Halides) to the eight properties that needed to be estimated. Ultimately, this dataset serves as the “textbook” for the algorithm before it embarks into unexplored chemical territory.

3. Basis of Prediction

What we are developing is a fully-fledged predictive engine that will help to fill the niche of connection between quantum properties on an atomic scale and engineering applications on a macro-scale level. In order to be considered a robust and powerful model, however, our predictive engine must not only perform well in one task but have ability to predict multiple physical parameters simultaneously.

Our goal is to develop the engine that will predict the following eight physical parameters. They include basic thermodynamic and structural measures (atomization energy, formation energy, unit cell lattice parameters) and complex electronic properties (bandgap, dielectric constant). As a result of successful prediction of these exact set of quantum properties and macroscopic properties, we are going to create a computational tool that can detect materials required for high-tech purposes such as energy storage.

4. Feature Engineering

However, for the successful application of machine learning in materials discovery, one has to solve an essential problem associated with the algorithmic nature of computers. Namely, computer programs cannot interpret atomic drawings – they can only operate on numerical data. Slide 6 solves this problem via the introduction of chemo-structural fingerprinting, which transforms the physical structure of the polymer chain into a strictly numeric form.

This involves mapping the candidate material into a 20-dimensional vector using a set of rigorous calculations based on the fraction of its components. We consider the fraction of monomers, dimers, and trimers, which represent single blocks, two-block sequences, and three-block sequences, respectively.

Thus, the fingerprint reflects the local chemical environment of the whole polymer chain. Such a sophisticated numeric representation allows the machine learning algorithm to process chemical similarity, identify the physical patterns behind them, and accurately predict macroscopic properties.

The second technique is based on a concept called the Hohenberg-Kohn theorem. This theory postulates that the ground-state electron density of a substance defines all of its properties. As opposed to the other two techniques that merely classified the atomic units, the algorithm analyzes the actual quantum "glue" keeping the atoms in place.

In order to convert the above property into digital information, scientists measured the one-dimensional planar average charge density for each polymer along the chains' axes. The problem with such a measurement was that the starting point could be placed anywhere along the chain, which affected the results. To overcome this issue, a Fourier transformation was applied to the density data. With the Fourier transformation, the frequencies inherent to the charge density could be measured, which created a fingerprint of the molecule in terms of its quantum characteristics rather than its structural parameters. This theory is based on DFT.

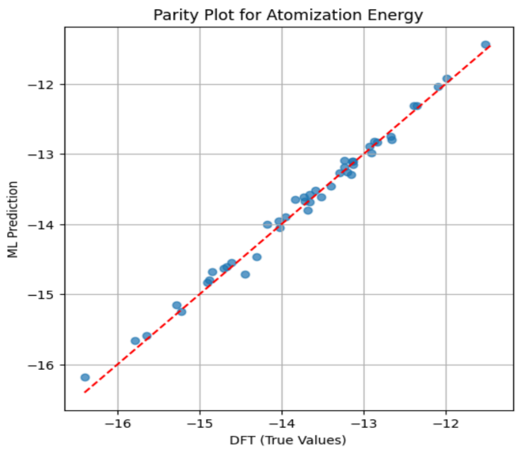
5. Kernel Ridge Regression(KRR)

We used the Kernel Ridge Regression method to map our number fingerprints to real properties. Not only does this algorithm learn from data, but it also computes the precise "chemical distance" of an unknown compound relative to the entire database that has been seen before. Our algorithm was built using 130 highly accurate quantum mechanical calculations and held back 45 for testing purposes.

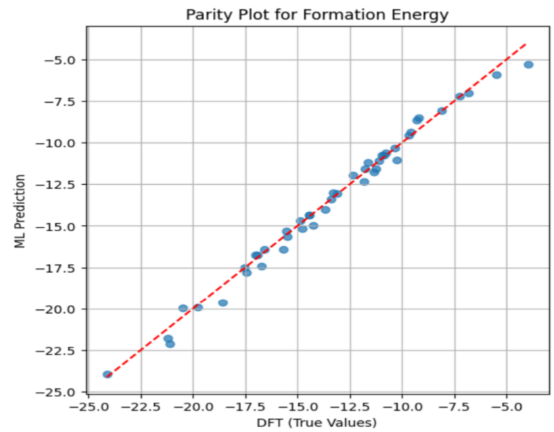
From a mathematical perspective, the Kernel Ridge Regression method uses an aggregate of weighted Gaussian functions to predict a given unknown compound's property. These functions are rescaled depending on the structural similarity of the number fingerprints between the test compounds and training samples. In order to avoid overfitting and actually learn the physics of matter instead of memorizing data points, we implemented five-fold rigorous cross-validation in our KRR engine.

RESULTS

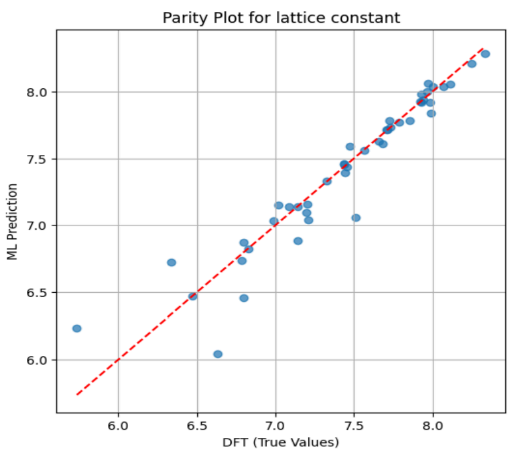
The performance of the Kernel Ridge Regression was assessed with the R2 score and the pearson correlation coefficient(r). We also plotted parity plots to visually examine the prediction quality.



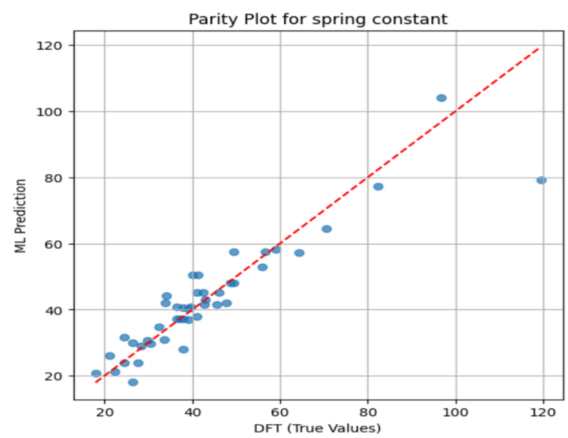
Pearson r: 0.9956102461544988



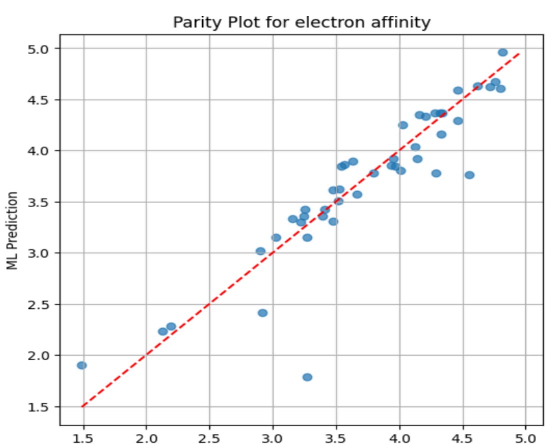
Pearson r: 0.9946530869792671



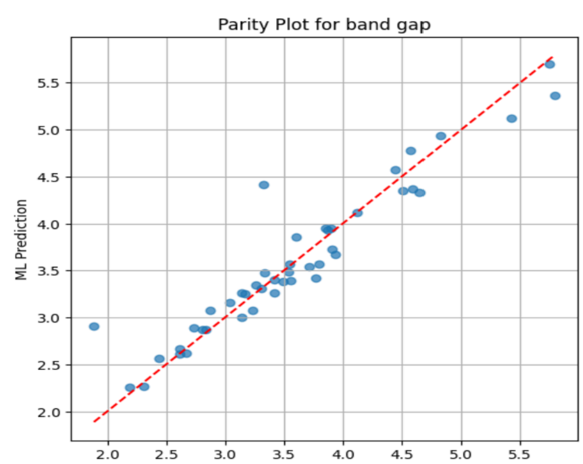
Pearson r: 0.9533724790918666



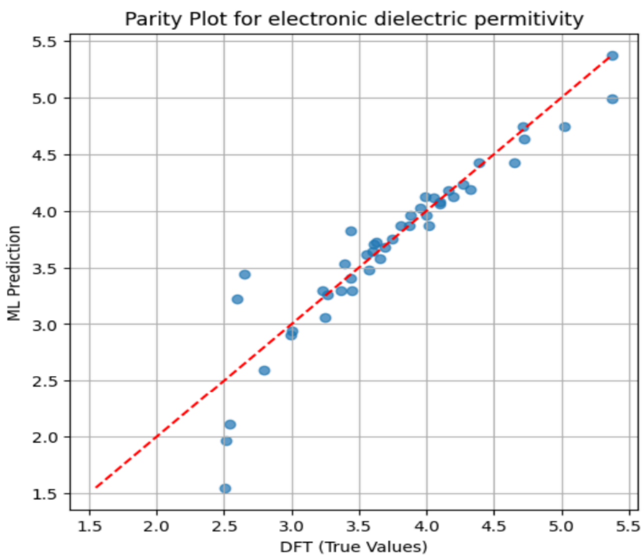
Pearson r: 0.9190620506762113



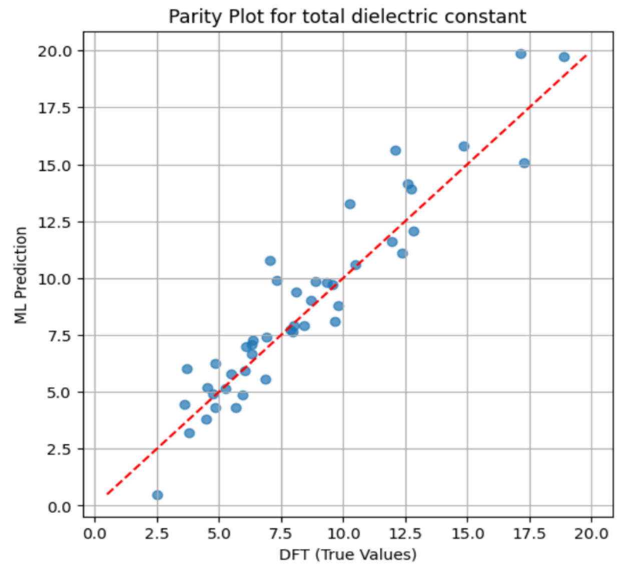
Pearson r: 0.9096194476890929



Pearson r: 0.9501089110488866



Pearson r: 0.9356851963683025



Pearson r: 0.945765670319672

Pearson correlation coefficient for the different models are:

Atomization Energy: 0.996

Formation Energy: 0.995

Lattice Constant: 0.954

Spring Constant: 0.919

Electron Affinity: 0.909

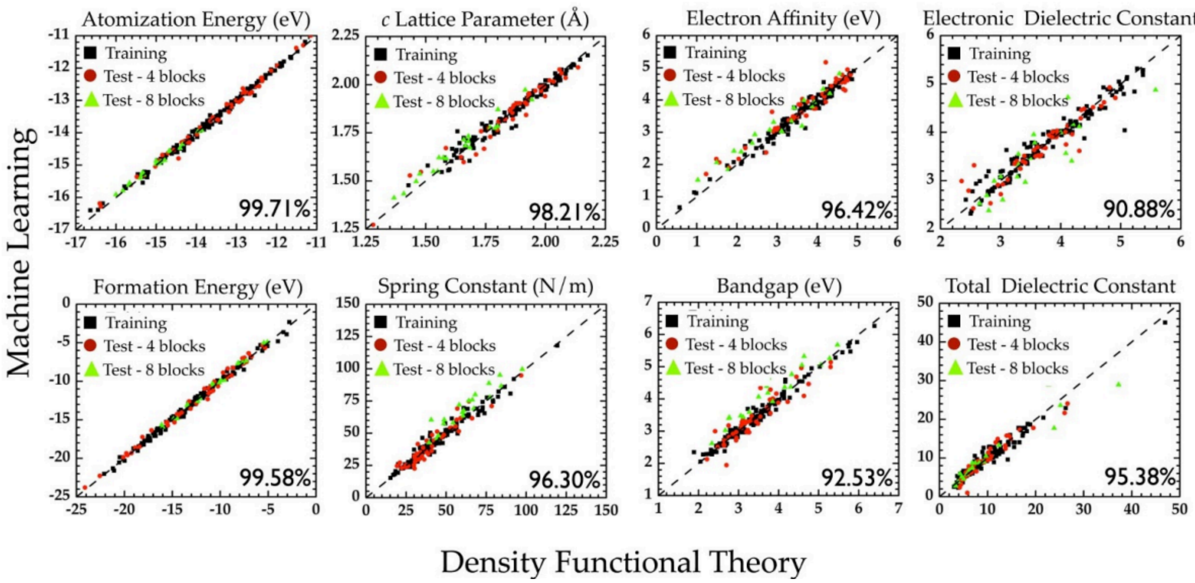
Band Gap: 0.950

Electronic Dielectric Constant: 0.936

Total Dielectric Constant: 0.946

The pearson Correlation coefficient was found to be greater tha 0.9 in all the models, indicating a strong linear relationship between the predicted values and the true values for all the properties.

The parity plots comparing the predicted values with the actual values also shows that data points cluster around the diagonal line in the case of all the properties, which indicates that the values predicted by the model are in strong agreement with the actual values. This is in agreement with the paper, Pilania et. al, whose results we attempted to reproduce, as can be seen in the graphs present in the paper.



As can be seen in the graphs from the paper, the data points cluster around the diagonal, similar to our results. The pearson correlation index, indicated as a percentage here, is also within 5 to 10 % of the value indicated in the paper.